

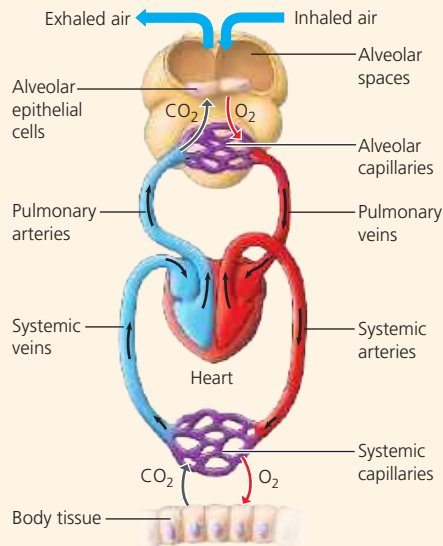
number and separation reflect adaptations to different environments and metabolic needs.

? How does the flow of a fluid in a closed circulatory system differ from the movement of molecules between cells and their environment with regard to distance traveled, direction traveled, and driving force?

CONCEPT 42.2

Coordinated cycles of heart contraction drive double circulation in mammals (pp. 926–929)

- The right ventricle pumps blood to the lungs, where it loads O_2 and unloads CO_2 . Oxygen-rich blood from the lungs entering the heart at the left atrium is pumped to the body tissues by the left ventricle. Blood then returns to the heart through the right atrium.



- The **cardiac cycle**, a complete sequence of the heart's pumping and filling, consists of a period of contraction, called **systole**, and a period of relaxation, called **diastole**. Heart function can be assessed by measuring the **pulse** (number of times the heart beats each minute) and **cardiac output** (volume of blood pumped by each ventricle per minute).
- The heartbeat originates with impulses at the **sinoatrial (SA) node** (pacemaker) of the right atrium. They trigger atrial contraction, are delayed at the **atrioventricular (AV) node**, and are then conducted along the bundle branches and Purkinje fibers, triggering ventricular contraction. The nervous system, hormones, and body temperature affect pacemaker activity.

? What changes in cardiac function might you expect after surgical replacement of a defective heart valve?

CONCEPT 42.3

Patterns of blood pressure and flow reflect the structure and arrangement of blood vessels (pp. 929–934)

- Blood vessels have structures well adapted to function. Capillaries have narrow diameters and thin walls that facilitate exchange. Arteries contain thick elastic walls that maintain blood pressure. Veins contain one-way valves that contribute to the return of blood to the heart.
- The velocity of blood flow is lowest in the capillary beds as a result of their large total cross-sectional area. Blood pressure is altered by changes in cardiac output and by variable constriction of arterioles.

- Fluid leaks out of capillaries and is returned to blood by the **lymphatic system**, which also defends against infection.

? If while standing you placed your forearm on your head, how, if at all, would the blood pressure in that arm change? Explain.

CONCEPT 42.4

Blood components function in exchange, transport, and defense (pp. 934–939)

- Whole blood consists of cells and cell fragments (**platelets**) suspended in a liquid matrix called **plasma**. Plasma proteins influence blood pH, osmotic pressure, and viscosity, and they function in lipid transport, immunity (antibodies), and blood clotting (fibrinogen). Red blood cells, or **erythrocytes**, transport O_2 . Five types of white blood cells, or **leukocytes**, function in defense against microorganisms and foreign substances in the blood. Platelets function in blood clotting, a cascade of reactions that converts plasma fibrinogen to fibrin.
- A variety of diseases impair function of the circulatory system. In **sickle-cell disease**, an aberrant form of **hemoglobin** disrupts erythrocyte shape and function, leading to blockage of small blood vessels and a decrease in the oxygen-carrying capacity of the blood. In cardiovascular disease, inflammation of the arterial lining enhances deposition of lipids and cells, resulting in the potential for life-threatening damage to the heart or brain.

? In the absence of infection, what percentage of cells in human blood are leukocytes?

CONCEPT 42.5

Gas exchange occurs across specialized respiratory surfaces (pp. 939–944)

- At all sites of **gas exchange**, a gas undergoes net diffusion from where its **partial pressure** is higher to where it is lower. Air is more conducive to gas exchange than water because air has a higher O_2 content, lower density, and lower viscosity.
- The structure and organization of respiratory surfaces differ among animal species. Gills are outfoldings of the body surface specialized for gas exchange in water. The effectiveness of gas exchange in some gills, including those of fishes, is increased by **ventilation** and **countercurrent exchange** between blood and water. Gas exchange in insects relies on a **tracheal system**, a branched network of tubes that bring O_2 directly to cells. Spiders, land snails, and most terrestrial vertebrates have internal **lungs**. In mammals, air inhaled through the nostrils passes through the pharynx into the **trachea**, **bronchi**, **bronchioles**, and dead-end **alveoli**, where gas exchange occurs.

? Why does altitude have almost no effect on an animal's ability to rid itself of CO_2 through gas exchange?

CONCEPT 42.6

Breathing ventilates the lungs (pp. 944–946)

- Breathing** mechanisms vary substantially among vertebrates. An amphibian ventilates its lungs by **positive pressure breathing**, which forces air down the trachea. Birds use a system of air sacs as bellows to keep air flowing through the lungs in one direction only, preventing the mixing of incoming and outgoing air. Mammals ventilate their lungs by **negative pressure breathing**, which pulls air into the lungs when the rib muscles and **diaphragm** contract. Incoming and outgoing air mix, decreasing the efficiency of ventilation.
- Sensors detect the pH of cerebrospinal fluid (reflecting CO_2 concentration in the blood), and a control center in the brain adjusts breathing